

* 3: linear transformations & matrices

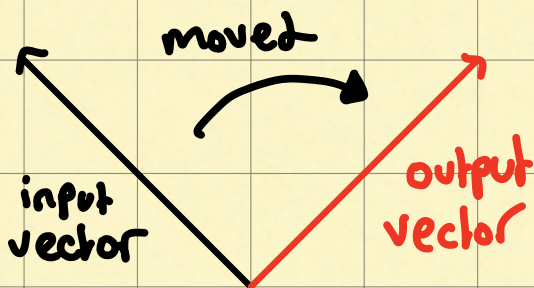
- linear transformation $\Rightarrow$ relation to matrices
- func

vector input

vector output

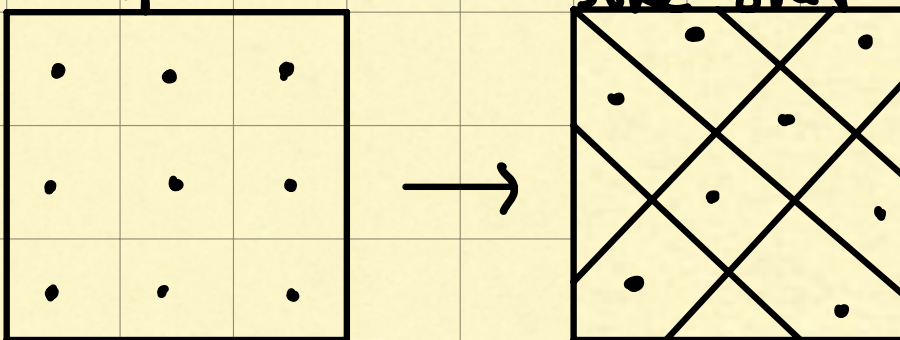
- transformation $\Rightarrow$ think using movement

ex:

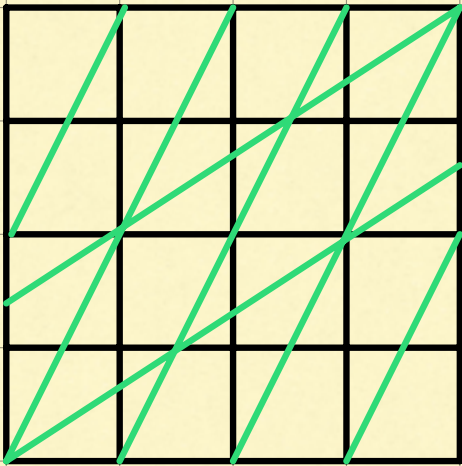


- to understand transformation as a whole:
imagine watching every possible input vector
move over to its corresponding output vector

- conceptualize each vector as a point where
its tip sits. then you'll see every point move to
some other point



- transformations in 2D: imagine points on an infinite grid. old grid is



$x_1 = \text{old}$
 $x_2 = \text{new}$
after
transformation

gives the feeling
of squishing & stretching
space itself

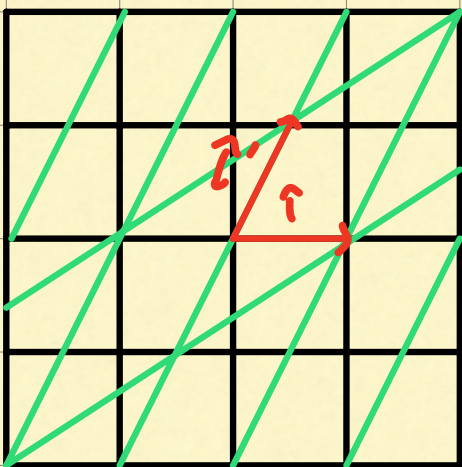
- linear algebra limits itself to a special type of transformation, easier to understand
- a transformation is linear if it has two properties
 - 1) all lines remain lines
 - 2) origin must remain fixed

→ keeps grid lines parallel & evenly spaced

- simple example: rotate grid abt the origin
- How to describe transformations numerically?

$$\begin{matrix} \begin{bmatrix} x_{in} \\ y_{in} \end{bmatrix} \\ \text{input vector} \end{matrix} \rightarrow ??? \rightarrow \begin{matrix} \begin{bmatrix} x_{out} \\ y_{out} \end{bmatrix} \\ \text{output vector} \end{matrix}$$

- you only need to record where basis vectors $\hat{i}$ & $\hat{j}$ land after the linear transformation & everything else will follow from that



example $\hat{i}$ became $\hat{i}'$

then: $\begin{bmatrix} -1 \\ 2 \end{bmatrix} = -1\hat{i} + 2\hat{j} = \vec{v}$

then $\vec{v}$ in new space after linear

transformation is still $\vec{v}' = -1\hat{i}' + 2\hat{j}'$

So you can find any vector after

linear transformation, if you have old & new $\hat{i}, \hat{j}$ vectors

- write vector w/ general coordinates $\begin{bmatrix} x \\ y \end{bmatrix}$

$$\hat{i} \rightarrow \begin{bmatrix} 1 \\ -2 \\ i' \end{bmatrix} \quad \hat{j} \rightarrow \begin{bmatrix} 3 \\ 0 \\ j' \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow x \begin{bmatrix} 1 \\ -2 \end{bmatrix} + y \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1x + 3y \\ -2x + 0y \end{bmatrix}$$

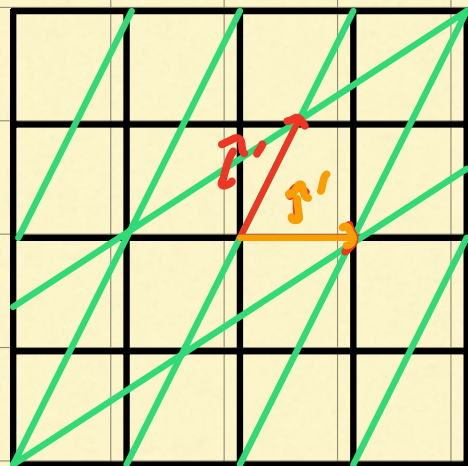
input vector

I give you any vector, & you can tell me where any vector lands using this formula

- a 2D linear transformation is completely described by, just 4 numbers
- $\hat{i}'$ coords & $\hat{j}'$ coords

$$\begin{matrix} \nearrow [?] \hat{j}' \\ \searrow \begin{bmatrix} 3 \\ -2 \end{bmatrix} \hat{i}' \end{matrix} \Rightarrow \begin{bmatrix} 3 & 2 \\ -2 & 1 \end{bmatrix}$$

where $\hat{i}$ lands where $\hat{j}$ lands



$$\cdot \begin{bmatrix} 5 \\ 7 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -2 & 1 \end{bmatrix} = 5 \begin{bmatrix} 3 \\ -2 \end{bmatrix} + 7 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

random
vector

linear
transformation

= adding scaled
versions of

our new basis
vectors

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

= matrix vector multiplication.

⇒ but they make u memorize
without showing high schoolers
the crucial part that makes
it feel intuitive

• It's more fun to think of columns as transformed part of basis vectors, & result as linear combination of them

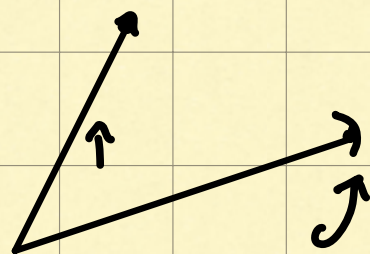
ex: 90° rotation counter clockwise

$\hat{i}$ lands on $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 $\hat{j}$ lands on $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\Rightarrow \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

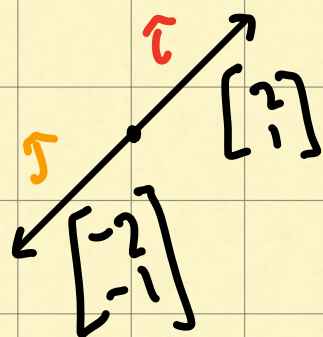
now you can see what happens to any vector after a 90° rotation

now how about the other way around?

$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \rightarrow$ deduce what it's transformation looks like



• if vectors $\hat{i}$ & $\hat{j}$ land on are linearly dependant (scaled versions of each other) it means $\begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$ the linear transformation squishes all of 2D space onto the line where they meet



(the 1D span of the two lin dep. vectors)

- lin transforms transform space s.t grid lines remain parallel & evenly spaced. These transformations can be described w/ just where each basis vector lands
- matrix columns = coords of basis vectors
- matrix-vector multip = compute what happens to vec.
- every time you see a matrix it's a translation of space.